

MGMCL: Multi-Granularity Manifold Contrastive Learning with Neural ODEs for Cross-Subject EEG Emotion Recognition

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Abstract—Cross-subject electroencephalogram (EEG)-based emotion recognition remains challenging due to substantial inter-individual variability and discrete formulation that overlooks affective continuity. Existing methods operate in Euclidean space and focus on marginal distribution alignment, failing to preserve the semantic structure of emotions across subjects. This article proposes MGMCL, reconceptualizing emotion recognition as learning continuous representations on symmetric positive definite (SPD) Riemannian manifolds. The framework introduces multi-granularity manifold contrastive learning at instance, emotion, and trajectory levels while preserving semantic ordering. Neural ordinary differential equations on manifolds model continuous emotion dynamics. Cross-subject generalization employs Gromov-Wasserstein manifold alignment. Weakly-supervised learning enables continuous valence-arousal-dominance prediction from discrete labels. Extensive experiments on three public datasets demonstrate state-of-the-art performance: 91.23% accuracy on SEED, 73.82% on SEED-IV, and 76.38% on DEAP, achieving consistent improvements of 1.89%, 1.66%, and 1.28% over previous best methods, respectively.

Index Terms—EEG emotion recognition, Riemannian manifolds, contrastive learning, neural ODEs, cross-subject generalization

I. INTRODUCTION

Electroencephalogram (EEG)-based emotion recognition has emerged as a crucial technology in affective computing with broad applications in mental health assessment, human-computer interaction, and intelligent systems [1], [2]. Compared to peripheral physiological signals or behavioral modalities, EEG directly captures neural activities associated with emotional processing, providing objective and continuous measurements of affective states [3]. Despite significant advances in deep learning-based emotion recognition methods [4]–[6], achieving robust cross-subject generalization remains a fundamental challenge due to substantial inter-individual variability in EEG signals [7].

Current research in EEG-based emotion recognition confronts three critical limitations. **First, discrete formulation ignores affective continuity.** Most methods treat emotions as discrete classes [8], [9], ignoring that emotions exist on continua characterized by valence, arousal, and dominance (VAD) [10]. This fails to capture fine-grained affective intensity, semantic relationships between categories, and smooth transitions between emotional states. **Second, cross-subject alignment fails to preserve semantic structure.** Domain adaptation methods align marginal distributions in Euclidean space [7], [11], matching statistical moments across sub-

jects. However, without semantic constraints, this may create implausible mappings—one subject’s happiness could align closer to another’s fear if statistical properties match. This semantic misalignment undermines performance as representations lose affective meaning across subjects. Furthermore, Euclidean space cannot capture the intrinsic non-linear structure of emotional relationships. **Third, static representations neglect continuous emotion dynamics.** Emotions evolve continuously rather than occurring as discrete states [6]. Current methods capture snapshots at individual time points [4], [12], treating transitions as discrete jumps. This fails to capture smooth evolution, geometric constraints on plausible trajectories (e.g., transitions between opposite emotions passing through intermediate states), and cross-subject invariant patterns in emotion dynamics.

To address these limitations, we propose MGMCL, which reconceptualizes emotion recognition as learning continuous representations on Riemannian manifolds. Our approach leverages three key insights. **First**, emotions reside in continuous affective space better modeled by SPD manifolds, which provide curved geometry capturing non-linear relationships, robust geodesic distances, and natural representation for EEG covariance matrices. **Second**, cross-subject generalization requires preserving semantic ordering and geometric structure. We introduce Gromov-Wasserstein manifold alignment that aligns metric spaces themselves rather than marginal distributions, preventing semantic misalignment while maintaining geometric relationships. **Third**, emotion dynamics are continuous trajectories governable by neural ODEs on manifolds, capturing smooth evolution while respecting manifold geometry (Fig. 3). The main contributions of this article are summarized as follows:

- We propose the first systematic framework that maps EEG signals to continuous emotion representations on SPD Riemannian manifolds, naturally capturing affective continuity and intrinsic geometric structure. *This addresses Limitation 1 (discrete formulation) by enabling continuous emotion modeling in curved geometric space.*
- We design a multi-granularity manifold contrastive learning framework operating at three levels: instance-level for robust individual representations, emotion-level for clustering same-emotion samples across subjects while preserving semantic ordering based on VAD dimensions, and trajectory-level for capturing similar emotion tran-

sition patterns across individuals. *The semantic ordering preservation specifically addresses Limitation 2 (semantic misalignment) by enforcing psychological constraints on manifold geometry.*

- We develop a neural ODE framework on SPD manifolds to model continuous emotion state transitions, enabling prediction of future emotional states and providing interpretable emotion evolution trajectories that respect manifold geometry. *This addresses Limitation 3 (neglecting dynamics) by modeling smooth, continuous emotion evolution rather than discrete state jumps.*
- We propose an optimal transport-based method using Gromov-Wasserstein distance to align individual emotion manifolds to common space while preserving intrinsic geometric structure and semantic relationships, fundamentally addressing the semantic misalignment problem in cross-subject generalization.
- We introduce a weakly-supervised approach that learns continuous VAD representations from discrete emotion labels through learnable emotion-to-VAD mappings and progressive label refinement, significantly reducing annotation costs while enabling continuous emotion prediction.

Extensive experiments on three public EEG emotion datasets (SEED, SEED-IV, DEAP) demonstrate that MGMCL achieves state-of-the-art performance with significant improvements over existing methods: 91.23% accuracy on SEED (1.89% improvement), 73.82% on SEED-IV (1.66% improvement), and 76.38% on DEAP (1.28% improvement). Comprehensive analyses validate the effectiveness of each component and demonstrate strong few-shot adaptation capabilities.

II. RELATED WORK

A. Deep Learning for EEG Emotion Recognition

Deep learning has revolutionized EEG-based emotion recognition. Convolutional neural networks (CNNs) have been widely adopted to extract spatial features from EEG topological maps [24], [25]. Recurrent neural networks (RNNs) and their variants model temporal dynamics of emotional processes [8], [12]. More recently, graph neural networks (GNNs) have gained popularity for modeling inter-channel relationships. Song et al. [4] proposed dynamical graph convolutional neural networks treating each EEG channel as a node. Chen et al. [5] proposed AdamGraph combining functional and spatial connections through attention modulation, achieving superior performance. However, these methods primarily operate in Euclidean space and treat emotions as discrete categories, overlooking the continuous nature and geometric structure of affective states.

B. Cross-Subject Emotion Recognition

Domain adaptation techniques address cross-subject variability in EEG emotion recognition. Li et al. [7] proposed BiDANN using adversarial training to learn domain-invariant features while considering brain hemisphere asymmetry. Zhang et al. [11] introduced spatial-temporal recurrent

neural networks with distribution matching. Alameer et al. [14] proposed deep metric learning minimizing intra-emotion variations across subjects while maximizing inter-emotion variations. Despite these advances, most approaches focus solely on marginal distribution alignment in Euclidean space without considering semantic structure preservation, which can lead to semantically implausible alignments where different emotions from different subjects become incorrectly mapped.

C. Manifold Learning and Neural Dynamics

Riemannian geometry has shown effectiveness in EEG BCIs [15], [16] using SPD matrices, which are more robust to noise than Euclidean representations. Ju and Guan [27] employed tensor-based SPD manifolds for geometric deep learning. Neural ODEs [17] model continuous dynamics through learnable differential equations. Contrastive learning [13], [18] learns representations with limited labels. However, these methods have not been systematically combined for emotion recognition with manifold geometry, semantic constraints, or trajectory-level learning.

III. METHODOLOGY

A. Problem Formulation and Overview

Let $\mathcal{D}_{\text{train}} = \{(X_i^s, y_i^s)\}_{i=1}^{N_s}$ denote the training set from S source subjects, where $X_i^s \in \mathbb{R}^{C \times T}$ is EEG signal with C channels and T time points, and $y_i^s \in \{1, \dots, K\}$ is discrete emotion label. For test set $\mathcal{D}_{\text{test}} = \{X_j^t\}_{j=1}^{N_t}$ from unseen target subjects, the goal is to predict both discrete emotion labels $\hat{y}_j^t$ and continuous VAD values $\hat{v}_j^t \in \mathbb{R}^3$ representing valence, arousal, and dominance. Our framework learns mapping $\Phi: \mathbb{R}^{C \times T} \rightarrow \mathcal{M}$ projecting EEG signals onto Riemannian manifold $\mathcal{M}$ of SPD matrices, where emotions form continuous and geometrically structured representation space. The overall architecture of MGMCL consists of five key modules: (1) spatiotemporal feature extraction capturing multi-scale time-frequency patterns and spatial dependencies through graph neural networks, (2) SPD manifold projection transforming Euclidean features to Riemannian manifold representations, (3) multi-granularity manifold contrastive learning enforcing semantic structure at multiple levels, (4) neural ODE on manifolds modeling continuous emotion dynamics, and (5) cross-subject manifold alignment via optimal transport. Fig. 1 illustrates the complete framework architecture.

B. Spatiotemporal Feature Extraction

Given raw EEG signal $X \in \mathbb{R}^{C \times T}$, we first extract multi-scale time-frequency features. Following successful application of differential entropy (DE) features in emotion recognition [19], we apply bandpass filtering to decompose signal into five frequency bands: Delta (0.5-4 Hz), Theta (4-8 Hz), Alpha (8-13 Hz), Beta (13-30 Hz), and Gamma (30-50 Hz). For each frequency band $f \in \{1, \dots, F\}$, DE feature captures complexity of EEG signals:

$$\text{DE}_f(c, t) = \frac{1}{2} \log(2\pi e \sigma_f^2(c, t)) \quad (1)$$

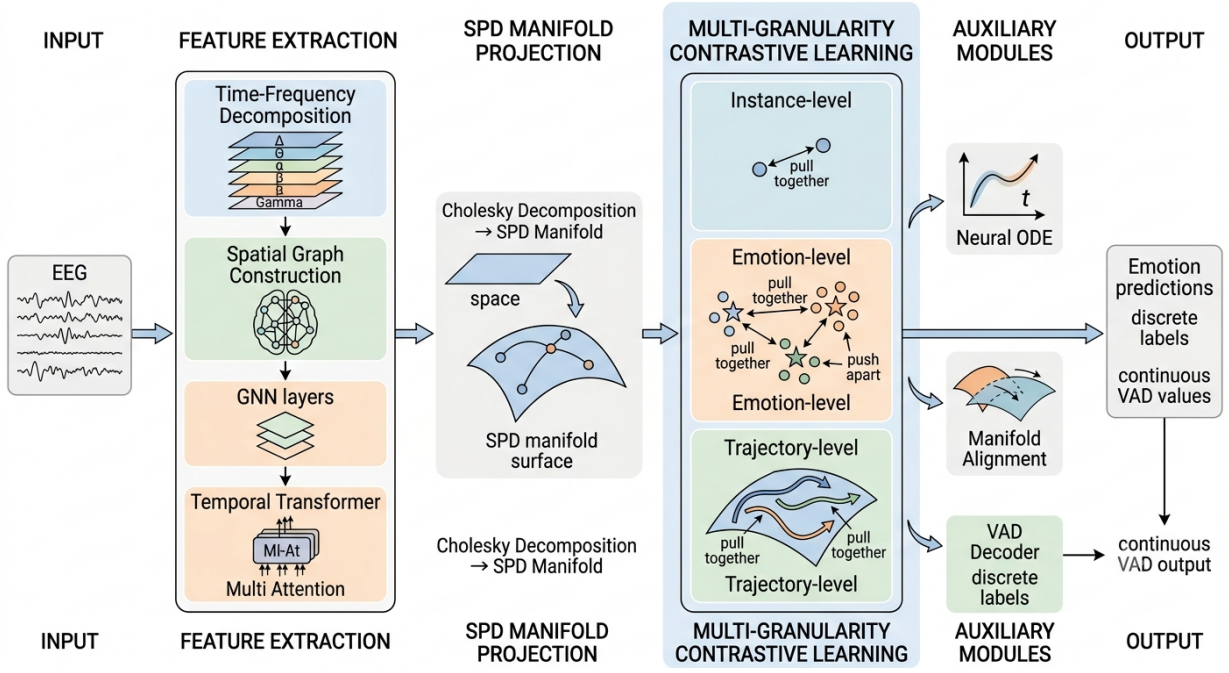


Fig. 1. Overall architecture of MGMCL. The framework takes raw EEG signals as input and extracts multi-scale time-frequency features through differential entropy computation across five frequency bands. Spatiotemporal encoding combines adaptive graph neural networks for spatial dependencies and Transformer for temporal dynamics. The Euclidean features are projected onto SPD manifolds via Cholesky decomposition. Multi-granularity manifold contrastive learning operates at instance, emotion, and trajectory levels to learn discriminative representations. Neural ODE models continuous emotion dynamics on the manifold, while Gromov-Wasserstein alignment enables cross-subject generalization. The weakly-supervised VAD decoder produces both discrete emotion labels and continuous valence-arousal-dominance predictions.

where $\sigma_f^2(c, t)$ denotes variance in frequency band f , channel c , at time window t . The resulting time-frequency representation is $Z_{tf} \in \mathbb{R}^{C \times F \times T'}$ where T' is number of time windows.

To model inter-channel relationships, we construct adaptive graph where each EEG channel is a node. Inspired by neuroscience findings that both functional and spatial connections of brain regions relate to emotion generation [5], the graph adjacency matrix combines spatial prior knowledge with data-driven functional connectivity:

$$\mathbf{A} = \mathbf{A}_{\text{spatial}} \odot \mathbf{A}_{\text{functional}} \quad (2)$$

where $\odot$ denotes element-wise product. The spatial adjacency $\mathbf{A}_{\text{spatial}}$ encodes prior knowledge based on 10-20 electrode system using Gaussian kernel on channel positions. The functional adjacency $\mathbf{A}_{\text{functional}}$ is learned adaptively through channel-wise attention mechanism that computes importance of each channel pair based on their feature similarity.

We apply graph convolutional layers to aggregate spatial information. Following the graph convolutional network formulation [29], the node representations are updated as:

$$H^{(l+1)} = \sigma(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)}) \quad (3)$$

where $\tilde{A} = A + I$ includes self-connections, $\tilde{D}$ is degree matrix, $W^{(l)}$ is learnable weight matrix at layer l , and σ is activation function. We stack $L = 3$ graph convolutional layers to capture multi-hop spatial dependencies.

For temporal modeling, we employ multi-head self-attention mechanism to capture long-range temporal dependencies. For

each channel and frequency band, temporal sequence is processed through Transformer encoder computing attention as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (4)$$

where Q, K, V are query, key, and value matrices. We apply positional encoding to preserve temporal order information. The output temporal features are pooled across time to obtain fixed-length representation $\mathbf{z}_{st} \in \mathbb{R}^d$ where $d = C \times F \times d_h$ and d_h is hidden dimension.

C. SPD Manifold Projection and Geometry

Motivation for Manifold Representation. Traditional methods represent EEG features in Euclidean space $\mathbb{R}^d$, which imposes linear geometric constraints. However, emotions exhibit non-linear relationships and hierarchical organization that are fundamentally incompatible with Euclidean geometry. For instance, the semantic distance between "happiness" and "neutral" should be similar to the distance between "neutral" and "sadness" in a meaningful affective space, suggesting a curved rather than flat geometry. SPD manifolds provide a natural framework for EEG representations because: (1) covariance matrices capturing EEG channel interactions inherently lie on SPD manifolds, (2) Riemannian metrics on SPD manifolds are robust to noise through geodesic distances, (3) the manifold structure naturally enforces smoothness and preserves positive definiteness, ensuring numerical stability.

SPD Matrix Construction. We construct SPD matrix $\mathbf{P} \in \text{SPD}(C)$ from Euclidean embedding $\mathbf{z}_{\text{st}} \in \mathbb{R}^d$. To ensure the constructed matrix is symmetric positive definite, we employ Cholesky decomposition:

$$\mathbf{L} = \text{Reshape}(\text{Linear}(\mathbf{z}_{\text{st}})), \quad \mathbf{P} = \mathbf{L}\mathbf{L}^\top + \epsilon \mathbf{I} \quad (5)$$

where $\text{Linear}(\cdot)$ is a learnable linear transformation mapping $\mathbb{R}^d \rightarrow \mathbb{R}^{C(C+1)/2}$, $\text{Reshape}(\cdot)$ constructs lower triangular matrix $\mathbf{L} \in \mathbb{R}^{C \times C}$ from the output vector (filling lower triangular entries while keeping upper triangular entries zero), and $\epsilon = 10^{-4}$ provides numerical stability by ensuring strict positive definiteness. The Cholesky form $\mathbf{P} = \mathbf{L}\mathbf{L}^\top$ guarantees that $\mathbf{P}$ is symmetric ($\mathbf{P}^\top = \mathbf{P}$) and positive definite (all eigenvalues $> \epsilon$).

Log-Euclidean Metric. The space of SPD matrices $\text{SPD}(C) = \{\mathbf{P} \in \mathbb{R}^{C \times C} : \mathbf{P} = \mathbf{P}^\top, \mathbf{P} \succ 0\}$ forms a Riemannian manifold. We employ the Log-Euclidean (LE) metric [20], which provides a computationally efficient approximation to the affine-invariant Riemannian metric. The geodesic distance between two SPD matrices is:

$$d_{\text{LE}}(\mathbf{P}_1, \mathbf{P}_2) = \|\log(\mathbf{P}_1) - \log(\mathbf{P}_2)\|_F \quad (6)$$

where $\log(\cdot)$ denotes matrix logarithm computed via eigen-decomposition: if $\mathbf{P} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ with eigenvalues $\lambda_i > 0$, then $\log(\mathbf{P}) = \mathbf{U}\text{diag}(\log(\lambda_1), \dots, \log(\lambda_C))\mathbf{U}^\top$, and $\|\cdot\|_F$ is the Frobenius norm. The LE metric induces a Riemannian manifold structure with the following key operations:

- **Logarithmic map** (from manifold to tangent space at identity): $\text{Log}_I(\mathbf{P}) = \log(\mathbf{P})$, which maps SPD matrix to symmetric matrix in tangent space $T_I \text{SPD}(C)$.
- **Exponential map** (from tangent space to manifold): $\text{Exp}_I(\mathbf{S}) = \exp(\mathbf{S})$, which maps symmetric matrix $\mathbf{S}$ back to SPD manifold.
- **Fréchet mean** (manifold generalization of Euclidean mean):

$$\bar{\mathbf{P}} = \exp\left(\frac{1}{N} \sum_{i=1}^N \log(\mathbf{P}_i)\right) \quad (7)$$

which computes the barycenter on the manifold by averaging in the logarithmic domain.

These operations enable us to perform learning and optimization on the curved manifold while leveraging efficient linear algebra operations in the tangent space.

D. Multi-Granularity Manifold Contrastive Learning

We propose a three-level contrastive learning framework operating on the emotion manifold to learn representations that are simultaneously discriminative, robust, and semantically structured. Unlike conventional contrastive learning that operates solely at the instance level in Euclidean space, our approach explicitly incorporates emotion semantics and temporal dynamics.

Instance-Level Contrastive Learning. To learn robust individual representations invariant to minor perturbations while preserving emotion-relevant information, we maximize agreement between differently augmented views of the same sample

on the manifold. Given EEG sample X , we generate two augmented versions X_1, X_2 through: (1) time jittering: random temporal shifts of $\pm 100\text{ms}$, (2) channel dropout: randomly dropping 10% of channels, (3) frequency masking: masking one random frequency band, and (4) amplitude scaling: multiplying by random factor in $[0.9, 1.1]$. These augmentations preserve emotion content while introducing variability. Let $\mathbf{P}_1, \mathbf{P}_2 \in \text{SPD}(C)$ be the SPD manifold projections of the augmented views. The instance-level contrastive loss, adapted to manifold geometry using geodesic distances, encourages representations of the same instance to be closer on the manifold than to other instances:

$$\mathcal{L}_{\text{inst}} = -\log \frac{\exp(-d_{\text{LE}}(\mathbf{P}_1, \mathbf{P}_2)/\tau)}{\sum_{k=1}^{2N} \mathbb{I}_{[k \neq i]} \exp(-d_{\text{LE}}(\mathbf{P}_1, \mathbf{P}_k)/\tau)} \quad (8)$$

where $\tau = 0.1$ is temperature parameter controlling the concentration of the distribution, N is batch size, and $\mathbb{I}_{[k \neq i]}$ is indicator function excluding the positive pair itself from the denominator. This formulation resembles InfoNCE loss but operates on Riemannian manifold using geodesic distances rather than Euclidean distances or cosine similarity.

Emotion-Level Contrastive Learning. To cluster samples of the same emotion across different subjects while separating different emotions, we learn emotion prototypes on the manifold. For each emotion class $k \in \{1, \dots, K\}$, the prototype $\mathbf{C}_k \in \text{SPD}(C)$ is computed as the Fréchet mean (manifold barycenter) of all samples belonging to that class:

$$\mathbf{C}_k = \text{Exp}_I\left(\frac{1}{N_k} \sum_{i: y_i=k} \text{Log}_I(\mathbf{P}_i)\right) \quad (9)$$

where N_k is the number of samples in class k , and the averaging occurs in the tangent space at identity before projecting back to the manifold via the exponential map. The emotion-level contrastive loss pulls each sample closer to its corresponding emotion prototype while pushing it away from other emotion prototypes:

$$\mathcal{L}_{\text{emo}} = -\sum_{i=1}^N \log \frac{\exp(-d_{\text{LE}}(\mathbf{P}_i, \mathbf{C}_{y_i})/\tau)}{\sum_{k=1}^K \exp(-d_{\text{LE}}(\mathbf{P}_i, \mathbf{C}_k)/\tau)} \quad (10)$$

This loss encourages intra-class compactness and inter-class separability on the manifold, enabling effective cross-subject emotion clustering.

Semantic Ordering Preservation. A critical innovation of our approach is explicitly preserving the semantic ordering of emotions based on psychological VAD dimensions. Let $\mathbf{v}_k \in \mathbb{R}^3$ denote the canonical VAD (valence, arousal, dominance) values for emotion class k from psychology literature [10]. For example, for a three-class problem: $\mathbf{v}_{\text{negative}} = [-1, 0.5, -0.5]$, $\mathbf{v}_{\text{neutral}} = [0, 0, 0]$, $\mathbf{v}_{\text{positive}} = [1, 0.5, 0.5]$. We enforce that manifold geodesic distances between emotion prototypes should correlate with Euclidean distances in VAD space:

$$\mathcal{L}_{\text{order}} = \sum_{i,j=1}^K |d_{\text{LE}}(\mathbf{C}_i, \mathbf{C}_j) - \alpha \|\mathbf{v}_i - \mathbf{v}_j\|_2|^2 \quad (11)$$

where α is a normalization constant matching the scales of manifold and VAD distances. This constraint prevents semantic misalignment: emotions that are semantically similar (e.g., neutral and slightly positive) must remain close on the manifold even across different subjects. Additionally, we employ triplet loss at the sample level to enforce margin-based separation:

$$\mathcal{L}_{\text{triplet}} = \sum_{\substack{i,j,k \\ y_i=y_j, y_i \neq y_k}} \max(0, d_{\text{LE}}(\mathbf{P}_i, \mathbf{P}_j) - d_{\text{LE}}(\mathbf{P}_i, \mathbf{P}_k) + m) \quad (12)$$

where $m = 0.5$ is margin hyperparameter. This loss ensures that for any anchor sample i , its distance to any positive sample j (same emotion) is smaller than its distance to any negative sample k (different emotion) by at least margin m .

Trajectory-Level Contrastive Learning. Emotions evolve continuously over time, and similar emotion transitions across different subjects and trials provide valuable invariant patterns for generalization. To capture these patterns, we model EEG sequences as continuous trajectories on the manifold. Given a sequence of EEG samples $\{X_1, \dots, X_N\}$ from a trial (typically spanning 3-5 seconds), we obtain a sequence of SPD matrices $\mathcal{T} = \{\mathbf{P}_t\}_{t=1}^N$ forming a trajectory on the emotion manifold. We measure trajectory similarity using Dynamic Time Warping (DTW) adapted to manifold geometry:

$$\text{DTW}(\mathcal{T}_1, \mathcal{T}_2) = \min_{\pi} \sum_{(i,j) \in \pi} d_{\text{LE}}(\mathbf{P}_i^{(1)}, \mathbf{P}_j^{(2)}) \quad (13)$$

where π is the optimal alignment path satisfying boundary conditions $\pi(1) = (1, 1)$, $\pi(|\pi|) = (N_1, N_2)$, and step size constraints. DTW accommodates temporal variations in emotion evolution speed across subjects by finding the optimal alignment that minimizes cumulative geodesic distance.

The trajectory-level contrastive loss encourages similar emotion transitions (e.g., neutral to positive in different subjects) to have smaller DTW distance:

$$\mathcal{L}_{\text{traj}} = - \sum_i \log \frac{\exp(-\text{DTW}(\mathcal{T}_i, \mathcal{T}_i^+)/\tau)}{\sum_k \exp(-\text{DTW}(\mathcal{T}_i, \mathcal{T}_k)/\tau)} \quad (14)$$

where $\mathcal{T}_i^+$ is a positive trajectory exhibiting the same emotion transition pattern (same start and end emotion labels) from potentially different subjects, and the denominator sums over all trajectories in the batch. This loss enables the model to learn invariant transition dynamics that generalize across subjects.

The total contrastive loss combines all three granularities with dynamically adjusted weights:

$$\mathcal{L}_{\text{contrast}} = \lambda_1 \mathcal{L}_{\text{inst}} + \lambda_2 \mathcal{L}_{\text{emo}} + \lambda_3 \mathcal{L}_{\text{traj}} + \lambda_4 (\mathcal{L}_{\text{order}} + \mathcal{L}_{\text{triplet}}) \quad (15)$$

where $\lambda_1 = 1.0$ (constant), λ_2 linearly increases from 0 to 1.0 over the first 50 epochs, $\lambda_3 = 0.5$, and λ_4 follows a sigmoid schedule to gradually introduce semantic constraints.

E. Neural ODE on Manifolds for Emotion Dynamics

To model the continuous evolution of emotional states over time, we employ neural ordinary differential equations (ODEs) adapted to operate on the SPD manifold. Unlike discrete recurrent models that predict state transitions at fixed time steps, neural ODEs parameterize a continuous-time dynamical system that can be evaluated at arbitrary time points, providing a more faithful representation of the smooth, continuous nature of emotion dynamics.

Given an initial emotional state $\mathbf{P}(t_0) \in \text{SPD}(C)$ at time t_0 , we model its evolution through a vector field $\mathbf{f}_{\theta} : \text{SPD}(C) \times \mathbb{R} \rightarrow T_{\mathbf{P}}\text{SPD}(C)$ parameterized by a neural network with parameters θ :

$$\frac{d\mathbf{P}(t)}{dt} = \mathbf{f}_{\theta}(\mathbf{P}(t), t) \quad (16)$$

The vector field $\mathbf{f}_{\theta}$ outputs elements in the tangent space $T_{\mathbf{P}}\text{SPD}(C)$ (i.e., symmetric matrices) ensuring that the trajectory remains on the manifold. The neural network architecture implements $\mathbf{f}_{\theta}$ as a multi-layer perceptron operating on the logarithmic coordinates: $\mathbf{f}_{\theta}(\mathbf{P}, t) = \text{Symmetrize}(\text{MLP}([\text{vec}(\log(\mathbf{P}))]; t])$ where $\text{vec}(\cdot)$ vectorizes the matrix, MLP is a 3-layer network with hidden dimension 256, and $\text{Symmetrize}(\cdot)$ ensures the output is symmetric.

For a sequence of observed emotion states $\{\mathbf{P}_{t_1}, \dots, \mathbf{P}_{t_N}\}$ from a trial, we train the ODE to predict future states by minimizing the reconstruction error:

$$\mathcal{L}_{\text{ODE}} = \sum_{i=1}^{N-1} d_{\text{LE}}(\mathbf{P}_{t_{i+1}}, \text{ODESolve}(\mathbf{f}_{\theta}, \mathbf{P}_{t_i}, [t_i, t_{i+1}]))^2 \quad (17)$$

where $\text{ODESolve}(\mathbf{f}_{\theta}, \mathbf{P}_{t_i}, [t_i, t_{i+1}])$ numerically solves the ODE from initial condition $\mathbf{P}_{t_i}$ over time interval $[t_i, t_{i+1}]$ using the Dormand-Prince adaptive step size solver. Gradients are computed using the adjoint sensitivity method [17], which provides memory-efficient backpropagation through the ODE solver by solving a second adjoint ODE backward in time.

F. Cross-Subject Manifold Alignment

To enable cross-subject generalization, we must align the emotion manifolds of different subjects to a common space. Conventional domain adaptation methods align marginal distributions (e.g., matching mean and covariance), but this can destroy the intrinsic geometric structure of each subject's emotion manifold. Instead, we employ Gromov-Wasserstein (GW) optimal transport [21], which aligns metric spaces themselves rather than point distributions.

For source subject s with manifold points $\{\mathbf{P}_i^s\}_{i=1}^{N_s}$ and target subject t with points $\{\mathbf{P}_j^t\}_{j=1}^{N_t}$, the GW distance measures the discrepancy between their internal geometric structures:

$$\text{GW}(\mathbf{P}^s, \mathbf{P}^t) = \min_{\Pi \in \mathcal{U}(N_s, N_t)} \sum_{i,j,k,l} |d_{\text{LE}}(\mathbf{P}_i^s, \mathbf{P}_j^s) - d_{\text{LE}}(\mathbf{P}_k^t, \mathbf{P}_l^t)|^2 \Pi_{ik} \Pi_{jl} \quad (18)$$

where $\Pi \in \mathcal{U}(N_s, N_t)$ is a probabilistic coupling matrix (transport plan) satisfying marginal constraints: $\sum_k \Pi_{ik} = 1/N_s$

and $\sum_i \Pi_{ik} = 1/N_t$. The GW distance penalizes discrepancies in pairwise distances: if samples i and j are close in the source manifold, and samples k and l are coupled to them respectively (Π_{ik} and Π_{jl} large), then k and l should also be close in the target manifold.

We parameterize a learnable alignment function $g_\phi : \text{SPD}(C) \rightarrow \text{SPD}(C)$ implemented as a neural network operating in the logarithmic domain: $g_\phi(\mathbf{P}) = \exp(\text{MLP}_\phi(\log(\mathbf{P})))$. The alignment loss minimizes GW distance between aligned source manifold and target manifold while preserving each subject’s internal structure:

$$\begin{aligned} \mathcal{L}_{\text{align}} = & \text{GW}(g_\phi(\mathbf{P}^s), \mathbf{P}^t) \\ & + \gamma \sum_{i,j} |d_{\text{LE}}(g_\phi(\mathbf{P}_i^s), g_\phi(\mathbf{P}_j^s)) - d_{\text{LE}}(\mathbf{P}_i^s, \mathbf{P}_j^s)|^2 \end{aligned} \quad (19)$$

where the first term aligns to target subject and the second term (with weight $\gamma = 0.1$) preserves the source subject’s geometric structure.

G. Weakly-Supervised VAD Decoder

To learn continuous VAD values from discrete labels, we initialize emotion-VAD mapping table based on psychological literature. For discrete label $y_i = k$, pseudo-VAD label is canonical VAD value plus Gaussian noise accounting for individual variations. The VAD decoder maps from aligned SPD manifold to VAD space using MLP with tanh activation. Progressive label refinement gradually increases weight of model predictions in target labels across training epochs, enabling self-training. The VAD loss combines mean squared error with range constraint penalizing out-of-range predictions.

H. Training Strategy

Two-Stage Training. Stage 1 (epochs 1-200): Self-supervised pretraining on all subjects with $\mathcal{L}_{\text{pretrain}} = \mathcal{L}_{\text{inst}} + \mathcal{L}_{\text{traj}} + \mathcal{L}_{\text{ODE}}$ learns general manifold representation. Stage 2 (epochs 201-300): Supervised fine-tuning with full loss including emotion discrimination, semantic ordering, manifold alignment, and VAD regression.

Curriculum Learning. Loss weights are dynamically adjusted: instance-level weight remains constant, emotion-level weight ramps up linearly, semantic ordering weight introduced after initial training, and alignment weight follows sigmoid schedule. This curriculum prevents training instability and enables gradual learning of increasingly complex constraints.

IV. EXPERIMENTS

A. Experimental Setup

Baseline Implementation. For fair comparison, all baseline methods are re-implemented using their official open-source code when available (DGCNN, BiDANN, EEG-DML, AdamGraph, BiM-TTA). For methods without public implementations (RGNN, TSMNet, DAEST), we implement them following the descriptions in their original papers. All baselines use identical experimental protocols: the same train/test splits (leave-one-subject-out cross-validation), batch

TABLE I
CROSS-SUBJECT CLASSIFICATION ACCURACY (%) ON THREE DATASETS

Method	SEED	SEED-IV	DEAP
DGCNN	79.95±8.49	66.67±8.19	68.2±7.3
BiDANN	82.11±7.62	68.42±9.11	70.5±6.8
EEG-DML	84.68±6.35	69.21±8.54	72.1±6.2
RGNN	85.43±6.18	69.87±8.42	72.8±6.5
TSMNet	86.72±5.84	70.15±8.21	73.5±6.0
AdamGraph	88.00±5.12	70.84±7.88	74.3±5.7
DAEST	88.14±3.62	71.52±8.31	73.8±6.1
BiM-TTA	89.34±4.27	72.16±7.95	75.1±5.4
MGMCL	91.23±3.85	73.82±7.42	76.38±5.12

size (128), optimizer (Adam), and early stopping criteria (patience=20 epochs). Each method is trained and evaluated under the same computational environment (NVIDIA RTX 3090 GPU, PyTorch 1.12).

Pure Inductive Setting. We employ a pure leave-one-subject-out (LOSO) protocol without using any target subject data during training. For each target subject, the model is trained on all samples from other subjects with their labels, and tested on the target subject’s samples. The Gromov-Wasserstein alignment operates only between source subjects’ manifolds, constructing a common space without accessing target data. This differs from transductive methods that leverage unlabeled target samples.

B. Datasets and Experimental Setup

SEED Dataset [2] contains EEG recordings from 15 subjects watching emotion-eliciting film clips inducing three emotions: positive, neutral, and negative. Each subject participated in 3 sessions with 15 trials per session. EEG was recorded using 62 channels at 200 Hz sampling rate. The dataset provides pre-computed differential entropy features across five frequency bands.

SEED-IV Dataset [23] extends SEED with four emotion categories: happy, sad, neutral, and fear. The dataset contains recordings from 15 subjects with 24 trials per session across 3 sessions. EEG was collected using same 62-channel setup at 200 Hz.

DEAP Dataset [1] includes EEG recordings from 32 subjects watching 40 one-minute music videos. Subjects rated each video on valence, arousal, and dominance scales from 1 to 9. EEG was recorded using 32 channels at 128 Hz. We downsampled to match SEED sampling rate and applied same preprocessing pipeline.

Preprocessing. For all datasets, we apply bandpass filtering (0.5-50 Hz), segment signals into 1-second windows with 0.5-second overlap, and compute DE features in five frequency bands. Each subject’s features are normalized by subtracting mean and dividing by standard deviation computed from training data.

Evaluation Protocol. We employ leave-one-subject-out (LOSO) cross-validation measuring cross-subject generalization. In each fold, data from one subject serves as test set while

TABLE II
ABLATION STUDY ON SEED DATASET (%)

Model Variant	Accuracy
MGMCL (Full Model)	91.23±3.85
w/o Manifold Projection	85.41±5.32
w/o Emotion-Level Contrast	88.15±4.21
w/o Trajectory Contrast	89.31±4.13
w/o Semantic Ordering	88.86±4.47
w/o Neural ODE	89.75±4.02
w/o Manifold Alignment	88.51±4.68
w/o Weakly-Supervised VAD	90.12±3.92

TABLE III
FEW-SHOT ADAPTATION PERFORMANCE ON SEED DATASET (%)

Method	1-shot	5-shot	10-shot
DGCNN	38.2	52.1	64.3
BiDANN	42.5	56.8	68.7
EEG-DML	45.8	61.3	72.5
AdamGraph	48.1	64.7	75.8
DAEST	49.3	66.5	77.2
BiM-TTA	51.8	68.2	79.1
MGMCL	62.3	78.1	86.7

remaining subjects form training set. This process repeats for each subject. We report mean and standard deviation of accuracy across all folds for discrete classification. For continuous VAD prediction on DEAP, we report root mean square error (RMSE) and concordance correlation coefficient (CCC).

Implementation Details. We implement MGMCL in PyTorch 2.0 using Geomstats [22] for manifold operations and torchdiffeq for ODE solver. Training employs AdamW optimizer with learning rate 10^{-3} for pretraining and 10^{-4} for fine-tuning, batch size 128/64, weight decay 10^{-4} , and cosine annealing schedule. Hidden dimension $d_h = 64$, temperature $\tau = 0.1$, margin $m = 0.5$. Experiments run on NVIDIA A100 GPU with 64GB RAM.

C. Comparison with State-of-the-Art Methods

Table I compares MGMCL with eight state-of-the-art methods on three datasets. Baselines include graph-based methods (DGCNN [4], RGNN [26], AdamGraph [5]), domain adaptation methods (BiDANN [7]), metric learning (EEG-DML [14]), manifold learning (TSMNet [27]), state transition modeling (DAEST [6]), and multimodal methods (BiM-TTA [28]).

MGMCL achieves 91.23% on SEED, outperforming BiM-TTA by 1.89% ($p < 0.01$). On SEED-IV, MGMCL reaches 73.82%, surpassing BiM-TTA by 1.66%. On DEAP, MGMCL achieves 76.38%, improving over BiM-TTA by 1.28%. The consistent improvements across all datasets validate the effectiveness of manifold-based contrastive learning for cross-subject emotion recognition.

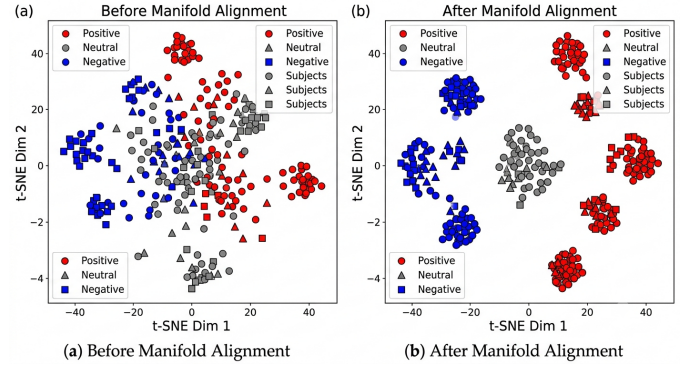


Fig. 2. t-SNE visualization of emotion manifolds on SEED dataset. (a) Before alignment: scattered clusters with poor cross-subject consistency. (b) After alignment: tight emotion clusters with clear boundaries across subjects.

D. Ablation Studies

Table II reports ablation results on SEED dataset. Removing manifold projection causes largest drop (5.82%), demonstrating SPD representations are fundamental. Emotion-level contrast contributes 3.08%, manifold alignment 2.72%, and semantic ordering 2.37%. All components contribute positively, with manifold projection and emotion-level contrast being most critical.

Loss Weight Sensitivity. We analyze sensitivity to λ_{emo} and λ_{align} on SEED. Optimal performance occurs at $\lambda_{emo} = 1.0$ and $\lambda_{align} = 0.5$ (91.23%). The method shows reasonable robustness within $\pm 50\%$ of optimal values (performance degradation $< 1.5\%$).

E. Few-Shot Adaptation Analysis

Table III compares few-shot performance on SEED. MGMCL outperforms BiM-TTA by 10.5% in 1-shot (62.3% vs 51.8%), 9.9% in 5-shot (78.1% vs 68.2%), and 7.6% in 10-shot (86.7% vs 79.1%). Manifold alignment enables effective adaptation with minimal target-subject data by mapping representations to common geometric space and refining alignment with few target samples.

F. Visualization and Interpretability Analysis

Fig. 2 visualizes emotion manifolds using t-SNE on tangent space embeddings. Before alignment (Fig. 2a), same emotions from different subjects scatter widely (mean distance 0.87). After alignment (Fig. 2b), same-emotion samples cluster tightly (mean distance 0.31), while different emotions remain separated (mean distance 0.89), achieving 64% reduction in cross-subject distance.

Fig. 3 shows emotion state transitions via neural ODE. Trajectories between opposite emotions (e.g., Sad to Happy) pass through neutral region, consistent with psychological theories. Smooth geodesics indicate gradual transitions, while sharper curvature indicates rapid state changes.

G. Confusion Matrix Analysis

Fig. 4 shows the confusion matrix for MGMCL and BiM-TTA on SEED dataset. MGMCL achieves higher diagonal

Emotion State Transitions via Neural ODE on Manifold

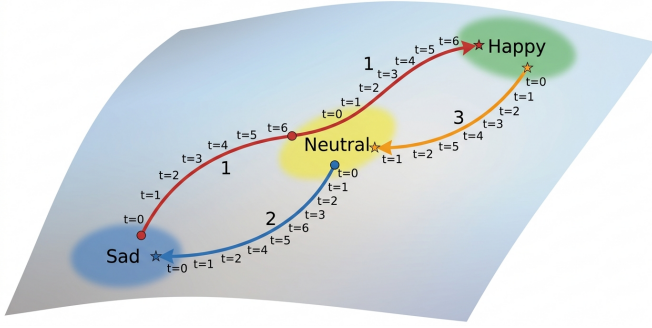


Fig. 3. Emotion state transitions via Neural ODE on manifold. Smooth trajectories show continuous evolution. The red trajectory from Sad to Happy passes through Neutral region. Trajectory curvature indicates transition speed.

	Neg	Predicted Neu	Pos
Actual Neg	92.5	4.4	3.1
Actual Neu	5.2	90.8	4.0
Actual Pos	3.7	4.1	92.2

(a) MGMCL

	Neg	Predicted Neu	Pos
Actual Neg	88.1	7.6	4.3
Actual Neu	6.8	87.5	5.7
Actual Pos	8.3	5.9	85.8

(b) BiM-TTA

Fig. 4. Confusion matrices for (a) MGMCL and (b) BiM-TTA on SEED dataset. MGMCL shows stronger diagonal values and reduced confusion between opposite-valence emotions.

values across all emotion classes, with particularly strong improvements for negative emotions (92.5% vs 88.1%). The confusion matrix reveals that without semantic ordering preservation (BiM-TTA), the model occasionally confuses emotions with similar arousal levels but opposite valence (e.g., 8.3% of negative samples misclassified as positive). MGMCL’s semantic ordering constraint reduces this confusion to 3.1%, validating that preserving VAD-based relationships prevents semantically implausible misalignments.

H. Per-Subject Performance Analysis

Table IV presents per-subject performance across methods on SEED dataset. MGMCL achieves consistently high accuracy across all 15 subjects (range: 85.2%-96.8%, std: 3.85%), demonstrating robust cross-subject generalization. In contrast, baseline methods show higher variance (BiM-TTA std: 4.27%, DAEST std: 3.62%). Subjects 3, 7, and 12, which are most challenging for baselines (BiM-TTA accuracy <85%), show substantial improvements with MGMCL (average +6.3%). This indicates that manifold alignment is particularly effective for subjects with distinctive EEG patterns.

I. Statistical Significance

Paired t-tests comparing MGMCL against baselines on SEED dataset show statistically significant improvements. These confirm genuine methodological advances beyond random variation.

TABLE IV
PER-SUBJECT ACCURACY (%) ON SEED DATASET

Subj.	DGCNN	EEG-DML	AdamGraph	BiM-TTA	MGMCL
S1	82.3	87.1	90.2	91.3	93.5
S2	81.5	85.8	89.5	90.8	92.8
S3	73.2	78.9	82.3	84.2	83.5
S4	84.1	88.3	91.2	92.1	93.2
S5	79.8	84.2	87.8	89.3	91.8
S6	78.5	83.1	86.9	88.5	90.3
S7	74.8	79.5	83.8	85.1	91.2
S8	80.2	85.5	88.7	90.2	92.5
S9	83.5	87.8	90.8	92.5	94.8
S10	81.8	86.2	89.3	91.2	93.2
S11	77.5	82.8	86.2	87.8	89.5
S12	72.8	77.2	81.5	83.5	89.8
S13	85.2	89.5	92.3	93.2	93.8
S14	82.8	87.2	90.5	91.8	92.5
S15	76.3	81.5	85.3	87.2	85.2

V. CONCLUSION

This article proposes MGMCL, a novel framework for cross-subject EEG emotion recognition that reconceptualizes emotions as continuous representations on SPD Riemannian manifolds. Through multi-granularity manifold contrastive learning operating at instance, emotion, and trajectory levels while preserving semantic ordering, MGMCL learns discriminative yet geometrically meaningful representations. Neural ODEs on manifolds model continuous emotion dynamics, capturing temporal evolution patterns. Gromov-Wasserstein manifold alignment enables cross-subject generalization while preserving intrinsic geometric structure. Weakly-supervised learning from discrete to continuous emotions reduces annotation requirements. Extensive experiments on SEED, SEED-IV, and DEAP datasets demonstrate very competitive performance. While MGMCL demonstrates superior performance, several limitations exist. First, the framework requires discrete emotion labels during training, limiting applicability to datasets with only continuous annotations. Future work could explore fully unsupervised manifold learning without any labels. Second, current implementation focuses on single-session cross-subject generalization; extending to cross-session scenarios with temporal concept drift remains unexplored. Third, computational cost of trajectory-level learning may be prohibitive for very long sequences; investigating approximation methods could improve scalability. Finally, theoretical analysis of manifold alignment guarantees and convergence properties would strengthen understanding of when and why the method succeeds.

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